

Retrieving, aligning and comparing sequences

Evolutionary Thinking — TA session

Written for **MEGA12** · every menu path below was checked click by click

Today you will pull real sequences out of GenBank without leaving MEGA, align them two different ways, translate them to protein, and finish with a pairwise distance matrix that says how far apart your species actually are. The worked example is **FOXP2**, but any gene you find more interesting will work just as well.

What you need

- **MEGA12**, installed and open, and an internet connection — MEGA fetches sequences itself.
- Somewhere to write things down. You will produce two distance matrices today and want them again next week, so keep your notes and save your alignments.
- Work in pairs if you like. Comparing two people's choice of sequences is half the lesson.

1 Finding sequences, and the papers behind them

GenBank holds publicly deposited DNA, RNA and protein sequences; PubMed holds the literature about them. Both use the same search syntax, and ten minutes learning it will save you a lot of scrolling for the rest of the degree.

You restrict a term to a particular field by putting the field in square brackets after it, and you combine terms with **AND**, **OR** and **NOT** :

Field	Example
[Author]	Paabo S[Author]
[Title]	coronavirus[Title]
[pdat]	2020[pdat] — publication date
[Organism]	"Homo sapiens"[Organism]

Warm-up. Put this into PubMed and see what comes back:

Zhao[Author] AND Wu F[Author] AND Yu[Author] AND coronavirus[Title] AND 2020[pdat]

Questions — work through these before opening MEGA

1. Find the paper describing the first human genome sequence — either the International Human Genome Sequencing Consortium's or Venter *et al.*'s, both 2001. Which journals published them, and how did the two projects differ in approach?
2. Find the high-coverage Denisovan genome published by Svante Pääbo's group in 2012. Who were the Denisovans, and what does “high coverage” buy you that low coverage does not?
3. How many nucleotide sequences does GenBank hold for dolphins (*Delphinidae*)? Any other family is fine if you prefer — use the Latin name.
4. Find the taxonomic position of dolphins, then compare the number of deposited sequences across a few dolphin species. Why are the counts so uneven? What does that tell you about what sequence databases actually sample?
5. Search for FOXP2. What is the protein, and why has it attracted so much attention?

2 Getting sequences into MEGA12

MEGA12 has a web browser built in, so the whole retrieval step happens inside the program and sequences land directly in your alignment.

Opening the alignment editor

From the main MEGA12 window, open **ALIGN** and choose **Edit/Build Alignment**.

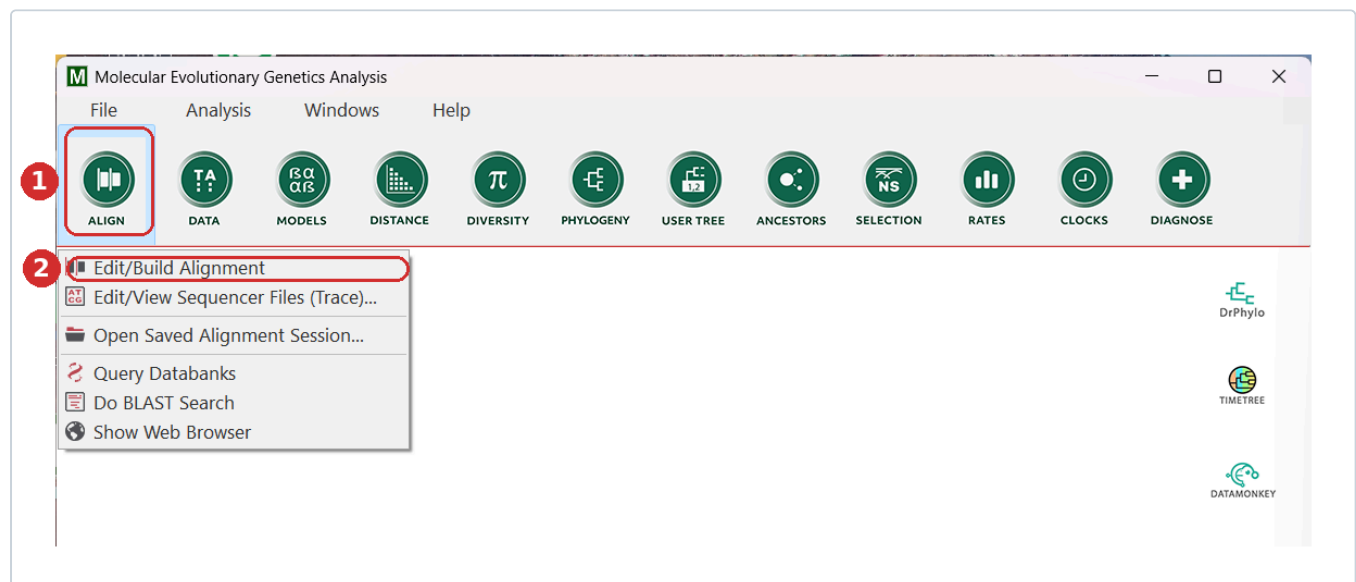


Figure 1. The MEGA12 main window. **1** the **ALIGN** button — **2** **Edit/Build Alignment**. This menu also carries **Query Databanks**, which opens the same GenBank browser you will use in a moment.

A small dialog asks what you want to do. Leave **Create a New Alignment** selected and press **Ok**.

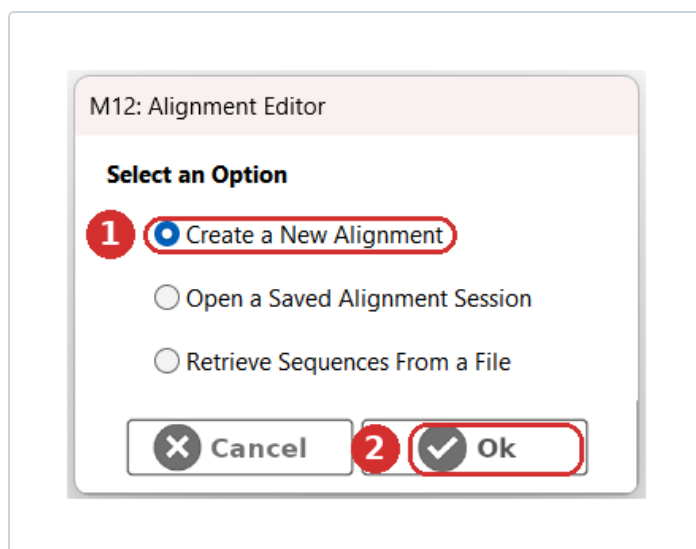


Figure 2. ① **Create a New Alignment** — ② **Ok**. Use **Retrieve Sequences From a File** instead if you already have a FASTA file to work from.

MEGA12 then asks whether you are building a DNA or a protein alignment. Choose **DNA**, and the Alignment Explorer window opens.

If you already had an alignment open, two extra confirmations appear first — “Are you creating a new data set?” (answer **Yes**) and “Save the current alignment session to file?” (answer **No** unless you want to keep it). On a freshly opened MEGA12 you will not see either.

Searching GenBank from inside MEGA

In the Alignment Explorer, open the **Web** menu and choose **Query Genbank**.

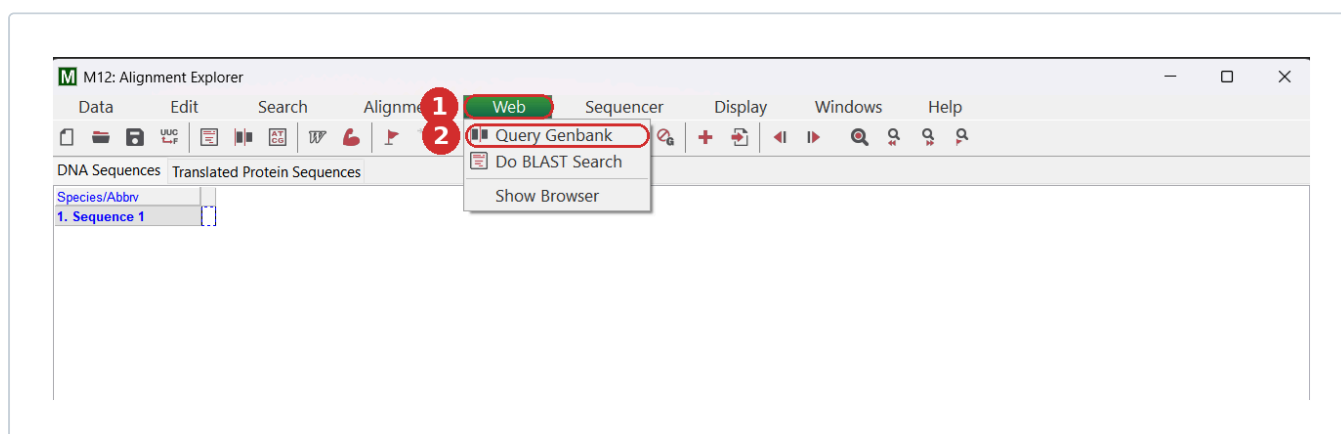


Figure 3. The Alignment Explorer. ① the **Web** menu — ② **Query Genbank**. **Do BLAST Search** directly underneath is the homology route described at the end of this section.

The built-in browser opens on the NCBI Nucleotide page. Use the search fields from section 1 to ask for exactly the kind of record you need. For the worked example:

```
FOXP2[Title] AND "complete cds"[Title] AND "Homo sapiens"[Organism]
```

The **"complete cds"[Title]** term is doing real work here: it throws away genomic contigs, partial fragments and non-coding transcripts, and leaves you with records that are actually the full coding sequence. That is the difference between an alignment you can translate and one you cannot — see the box on the next page.

Nucleotide

Nucleotide [v] FOXP2[Title] AND "complete cds"[Title] AND "Homo sapiens"[Organism]

Create alert Advanced

Species
Animals (6)
Customize ...

Molecule types
mRNA (6)
Customize ...

Source databases
INSDC (GenBank) (6)
Customize ...

Sequence Type
Nucleotide (6)

Sequence length
Custom range...

Release date
Custom range...

Revision date
Custom range...

[Clear all](#)

[Show additional filters](#)

Format

- ☒ Summary
- ☐ GenBank
- ☐ GenBank (full)
- ☐ FASTA
- ☐ ASN.1
- ☐ Revision History
- ☐ Accession List
- ☐ GI List

20 per page Sort by Default order Send to:

- Homo sapiens clone **FOXP2**_iso-C_fetal-F01 forkhead box P2 isoform C (**FOXP2**) mRNA, complete cds, alternatively spliced
2,358 bp linear mRNA
Accession: KJ535079.1 GI: 608785777
[Protein](#) [PubMed](#) [Taxonomy](#)
[GenBank](#) [FASTA](#) [Graphics](#)
- Homo sapiens brain forkhead/winged helix transcription factor **FOXP2** isoform mRNA, complete cds; alternatively spliced
2,594 bp linear mRNA
Accession: AY144615.1 GI: 24496248
[Protein](#) [Taxonomy](#)

The page turns into plain FASTA text. Check the header line is the record you meant to take, then press **+ Add To Alignment** in the browser toolbar and the sequence drops into your Alignment Explorer.

Figure 5. 1 + Add To Alignment. The header confirms this is KJ535079.1, human FOXP2, *complete cds* — the sequence in view is the one that gets added.

Now repeat for the other species. Keep the same search and swap only the organism, so that every sequence you collect is the same kind of record:

```
... AND "Pan troglodytes"[Organism] · ... AND "Gorilla gorilla"[Organism]
```

Three or four species is plenty. The worked example below uses human, chimpanzee and gorilla.

Choosing sequences — this is where alignments go wrong

- Use the **CDS** (coding sequence), and use the same kind of record for every species. If you take a CDS for human and a full mRNA for chimp, your alignment will be mostly gaps — and the gaps will be telling you about your record choice, not about evolution. Restricting the search with `"complete cds"[Title]` is the easiest way to keep yourself honest.
- Sequence lengths should be roughly comparable before you start.
- If only a genomic RefSeq entry exists, open it in GenBank view, find the **CDS** feature, and use its coordinates to pull out the coding part.
- Long genomic sequences align slowly. If MUSCLE seems to hang, check how many bases you gave it.

The BLAST route (alternative)

Instead of searching by name, you can start from one sequence and let homology find the rest. Click your sequence in the Alignment Explorer so it turns yellow, then **Web** →

Do BLAST Search. Your sequence should already be in the query box — if it is empty, you did not have the sequence selected. Scroll down, press **BLAST**, then on the results page clear **select all**, tick only the hits you want, and follow the GenBank links back to retrieve them exactly as above.

3 Aligning your sequences

Select everything with **Ctrl+A**, then open the **Alignment** menu. MEGA12 gives you four aligners.

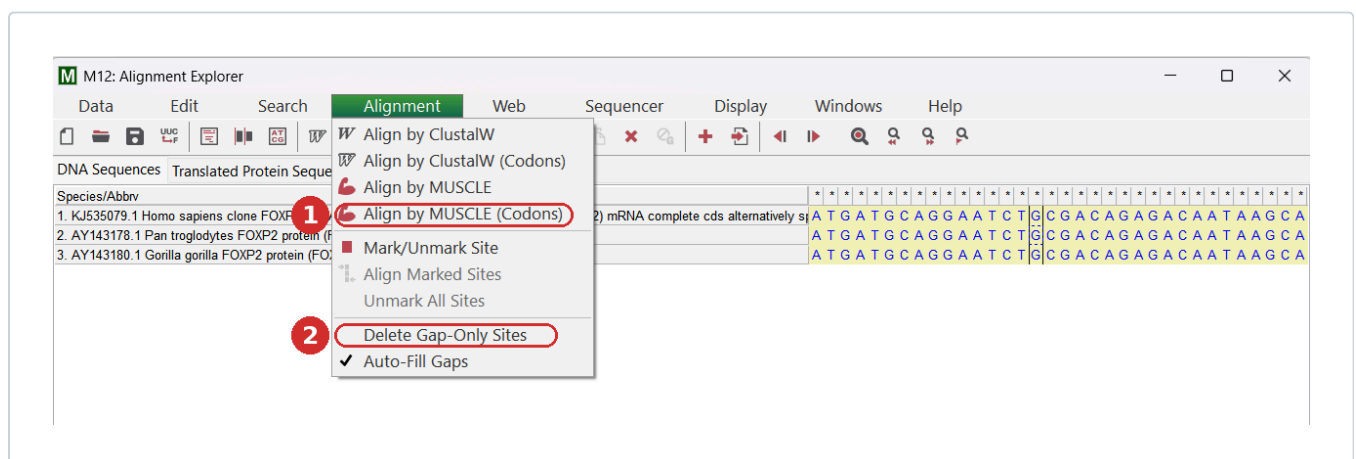


Figure 6. The three retrieved sequences, selected and ready. **1 Align by MUSCLE (Codons)** — the right choice for a coding sequence. **2 Delete Gap-Only Sites** is how you undo an alignment and start again.

The plain **ClustalW** and **MUSCLE** options treat your sequences as undifferentiated DNA. The **(Codons)** versions translate to protein, align there, and map the result back onto the DNA.

Why (Codons) matters for a CDS

A plain DNA aligner is free to open a gap of any length, including one or two bases. In a coding sequence that shifts the reading frame and produces protein that never existed. A codon-aware aligner can only ever insert gaps in multiples of three, so the frame survives. If you are working with a CDS, start with a **(Codons)** version — then run the plain one too and compare, because seeing the difference is the point of the exercise.

An options dialog appears. The defaults are reasonable; you can start by accepting them. Press **Ok** to start the alignment.

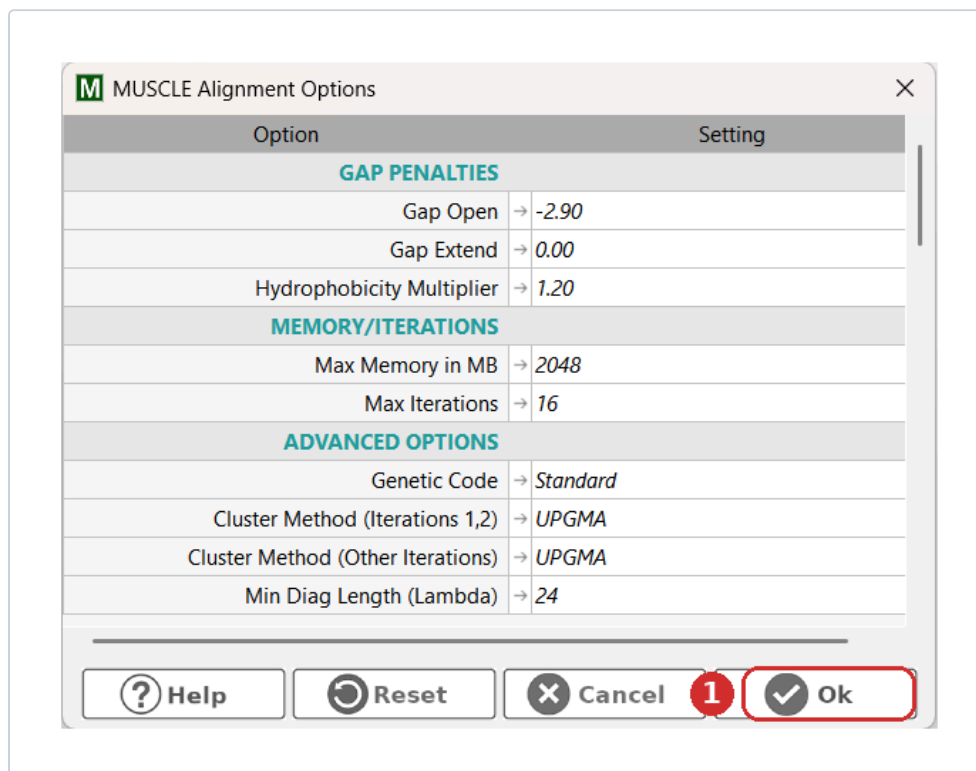


Figure 7. MUSCLE options. ① **Ok** starts the alignment. Come back to the gap penalties later if you want to see how much they change your result.

MEGA then asks “*Would you like to remove gaps before alignment?*”. Answer **Yes** if you are re-aligning sequences that are already aligned. MUSCLE runs in a separate window that sits on top of the Alignment Explorer — it is working, not frozen.

Questions — look at your alignment before moving on

1. How many gaps did you get, and where are they — scattered, or concentrated at the ends? What would each pattern mean?
2. Re-align with a different aligner or different gap penalties. What changed? Is one result obviously better, or just different?
3. Compare a codon-aware alignment with a plain DNA one on the same sequences. Do the gaps fall in the same places? Are any of them not a multiple of three?

From DNA to protein

Switch to the **Translated Protein Sequences** tab at the top of the Alignment Explorer. Check the genetic code shown in the bottom-right corner of the window is the one you want — change it under **Data → Genetic Code** if not.

Align the protein sequences too, and ask yourself which alignment is cleaner and why. A protein alignment of the same gene is almost always tidier — the interesting question is what that tells you about which changes selection actually sees.

When you are happy, export with **Data → Export Alignment → MEGA Format** to get a **.meg** file. Save both your DNA and your protein alignment — you will want them next week.

4 Pairwise distances

An alignment tells you which positions correspond. A distance matrix turns that into numbers. Go back to the main MEGA12 window and open **DISTANCE → Compute Pairwise Distances...**.

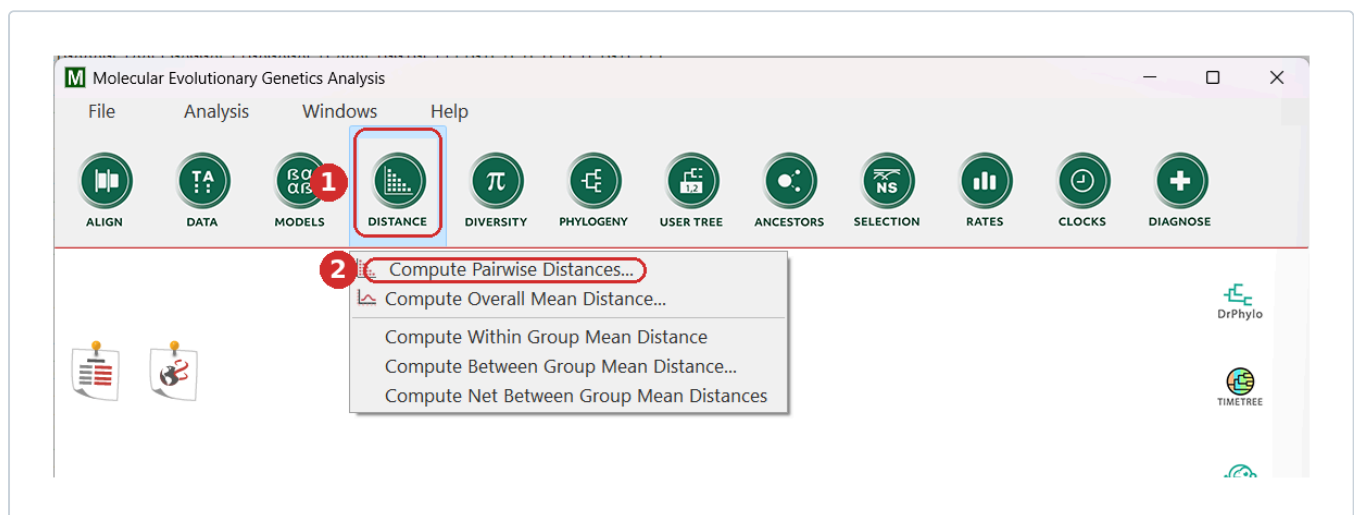


Figure 8. 1 the **DISTANCE** button — 2 **Compute Pairwise Distances...**, the first entry in the menu.

MEGA12 asks “Would you like to use the currently active data?” — press **Ok**. If nothing is loaded it will ask you to pick a file instead; give it the **.meg** file you just exported.

The Analysis Preferences window opens. The one setting you must change is **Model/Method**: set it to **p-distance**, which is simply the proportion of sites that differ. MEGA12 opens with a substitution model selected instead — and which one depends on what you last used — so **check this every time rather than trusting what is already in the box**.

M12: Analysis Preferences

Distance Estimation

Option	Setting
ANALYSIS	
Scope	Pairs of taxa
ESTIMATE VARIANCE	
Variance Estimation Method	None
SUBSTITUTION MODEL	
Substitutions Type	Nucleotide
Model/Method	Tamura-Nei model
Substitutions to Include	d: Transitions + Transversions
RATES AND PATTERNS	
Rates among Sites	Gamma Distributed (G)
Gamma Parameter	1.00
Pattern among Lineages	Same (Homogeneous)
DATA SUBSET TO USE	
Gaps/Missing Data	Pairwise deletion

Help Cancel Ok

Figure 9. Analysis Preferences. **1 Model/Method** — click the value and change it to **p-distance**. **2 Ok** runs the calculation. The model shown here is Tamura-Nei; yours may open on a different one.

Run this for your DNA alignment and again for your protein alignment, and write both matrices down.

Questions — the point of the whole session

1. Which pair of species is closest, and which is furthest? Is that what you expected from what you know about the phylogeny?
2. Your protein distances should be much smaller than your DNA distances for the same species. Why? What does the gap between the two numbers measure?
3. If two species have a protein distance of exactly zero but a non-zero DNA distance, what has been happening to that gene?
4. Try p-distance and then a substitution model such as Jukes-Cantor on the same alignment. Which distances are larger, and why should a model ever give a *bigger* number than simply counting differences?
5. Sketch a tree in which branch lengths are proportional to your distances. Can you do it so that every pairwise distance in your matrix is respected? What does it mean if you cannot?

5 If you finish early

- Swap FOXP2 for a gene you actually care about — haemoglobin across mammals, or a SARS-CoV-2 gene sampled in different countries — and run the whole pipeline again. The second pass is where it starts to feel routine.
- Add a distant outgroup and watch what it does to your distance matrix.
- Look at **nextstrain.org/ncov/global** for what this looks like at scale: the same logic, millions of sequences, updated live.